

DAHYE KIM

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RESEARCH INTERESTS

Image and Video Synthesis; Diffusion Models; Multimodal Models; Video Understanding; Interpretability

EDUCATION

Boston University <i>PhD Student in Computer Science</i> • Advisor: Prof. Deepti Ghadiyaram	MA, United States <i>Sep. 2024 - Current</i>
Yonsei University <i>M.S. of Electrical & Electronic Engineering</i> • Advisor: Prof. Kwanghoon Sohn	Seoul, South Korea <i>Sep. 2022 - Aug. 2024</i>
Yonsei University <i>B.S. of Electrical & Electronic Engineering</i> • Cumulative GPA: 4.0/4.0 (Class Rank: 2/224)	Seoul, South Korea <i>Mar. 2020 - Aug. 2022</i>
Hongik University <i>Electronic & Electrical Engineering</i>	Seoul, South Korea <i>Mar. 2018 - Feb. 2020</i>

PREPRINTS

1. **Dahye Kim**, Bhuvan Sachdeva, Karan Uppal, Naman Gupta, Vineeth N. Balasubramanian, and Deepti Ghadiyaram, “Swift Sampling: Selecting Temporal Surprises via Taylor Series,” *arXiv preprint arXiv:2605.22678*, 2026.
2. **Dahye Kim** and Deepti Ghadiyaram, “Concept Steerers: Leveraging K-Sparse Autoencoders for Test-Time Controllable Generations,” *arXiv preprint arXiv:2501.19066*, 2025.

PUBLICATIONS

1. **Dahye Kim**, Deepti Ghadiyaram, and Raghudeep Gadde, “DDiT: Dynamic Patch Scheduling for Efficient Diffusion Transformers,” In *CVPR*, 2026 (Highlight).
2. **Dahye Kim***, Xavier Thomas*, and Deepti Ghadiyaram, “*Revelio*: Interpreting and leveraging visual semantic information in diffusion models,” In *ICCV*, 2025; *CVPRW MIV*, 2025 (Oral).
3. **Dahye Kim**, Jungin Park, Jiyoung Lee, Seongheon Park, and Kwanghoon Sohn, “Language-free Training for Zero-shot Video Grounding,” In *WACV*, 2023 (Oral).
4. Seongheon Park, Hanjae Kim, Minsu Kim, **Dahye Kim**, and Kwanghoon Sohn, “Normality Guided Multiple Instance Learning for Weakly Supervised Video Anomaly Detection,” In *WACV*, 2023 (Oral).

RESEARCH EXPERIENCES

Boston University <i>Research Assistant (Advisor: Prof. Deepti Ghadiyaram)</i> • Generative Models; Multimodal Models; Interpretability	MA, United States <i>Sep. 2024 - Current</i>
Adobe <i>Research Scientist Intern (Mentor: Connelly Barnes)</i> • Efficient Diffusion Models	WA, United States <i>Aug. 2026 - Current</i>
Amazon <i>Applied Scientist Intern (GenAI team, Mentor: Raghudeep Gadde)</i> • Efficient Diffusion Models	WA, United States <i>Jun. 2025 - Nov. 2025</i>

Digital Image Media Lab., Yonsei University

Research Assistant (Advisor: Prof. Kwanghoon Sohn)

- Multi-modal Learning with Visual and Textual Representation

Seoul, South Korea

Sep. 2022 – Jul. 2024

Digital Image Media Lab., Yonsei University

Undergraduate Research Assistant (Advisor: Prof. Kwanghoon Sohn)

- Human-object Interaction (HOI) Detection; Video Understanding

Seoul, South Korea

Jan. 2021 - Aug. 2022

AWARDS AND HONORS

Honors

- High Honors Student (Top 3% of department)
- Honors Student (Top 10% of department)

Fall 2020, Spring 2021, Fall 2021

Spring 2020

Scholarships

- Brain Korea 21 (BK21) Scholarship, National Research Foundation of Korea
- Automotive System IC Scholarship,
Automotive System IC Fusion Human Resource Research Center
- Teaching Assistant Scholarship, Yonsei University (\$ 3,500 total)
- Graduate Student Research Assistant Scholarship, Yonsei University (\$ 3,500)
- Academic Excellence Scholarship, Yonsei University (\$ 3,600 total)
- National Science and Technology Scholarship (\$ 3,600),
National Research Foundation of Korea
- Academic Excellence Scholarship, Hongik University (\$ 6,300 total)

Sep. 2022 - July. 2024

Sep. 2022 - July. 2024

Fall 2022, Fall 2023

Fall 2022

Spring 2021, Fall 2021

Fall 2020

Fall 2018, Spring 2019, Fall 2019

PROFESSIONAL SERVICES

Workshop Organizer

- XAI4CV: Explainable Artificial Intelligence for Computer Vision (CVPR 2025, CVPR 2026)

Student Volunteer

- The 3rd New England Mechanistic Interpretability (NEMI) Workshop, Boston University

Aug. 2026

Reviewer

- Mechanistic Interpretability for Vision (MIV) at CVPR 2025
- AI for Content Creation (AI4CC) at CVPR 2026
- IJCV
- CVPR 2026
- ECCV 2026
- NeurIPS 2026
- AAAI 2027

TEACHING EXPERIENCES

CS 541: Applied Machine Learning

Spring 2025

EEE 3313: Introductory Digital Labs

Fall 2023

EEE 2060: Signals and Systems

Fall 2022

SELECTED COURSEWORK

Artificial Intelligence: Introduction Artificial Intelligence (A), Intelligent Control (A), Multimodal Deep Learning (A), Digital Image Processing (A), Medical Imaging System and its Application for Artificial Intelligence (A), Topics in Computer Vision (A), and Special Topics for Deep Learning (A)

Mathematics: Linear Algebra (A), Mathematical Statistics (A), Probability and Random Variables (A), Random Process (A), and Information Theory (A)

Programming: Data Structure and Algorithms (A), and Operating Systems (A)

OTHER SKILLS

Language and Tools: Python, MATLAB, C/C++, PyTorch, Linux Shell, LaTeX

Language Ability: English (fluent), Korean (native)